# Econometria Avançada — MPE (Insper)

Exercise list: *L01 — Review of basic ingredients (book exercises)*

*Solutions*

**Instructions.** Every question is a textbook exercise from the *Practice* list of Class 1 in `Map/class-plan.md`, reproduced verbatim from the book PDF (statements from Portuguese-language books stay in Portuguese). Show every step. Computer exercises are solved in Python; the script `Listas/L01-book-exercises-check.py` reproduces and asserts every number printed in the solutions. Book abbreviations follow the class plan: T = Tsay, W = Wooldridge, G = Greenberg, I = ISLP, B = Bolfarine & Sandoval, MS = Morettin & Singer.

## Block 1 — Tsay, *Analysis of Financial Time Series* *(T 2.2)*

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**Question 1.** T 2.2 — AR(2) moments and forecasts.

**Source:** Tsay, R. S. (2010), *Analysis of Financial Time Series*, 3rd ed., Wiley, Chapter 2 Exercises, Exercise 2.2, p. 104 — reproduced verbatim. (Header of the exercise set: "If not specifically specified, use 5% significance level to draw conclusions in the exercises.")

2.2. Suppose that the daily log return of a security follows the model

$$r_t = 0.01 + 0.2r_{t-2} + a_t,$$

where $\{a_t\}$ is a Gaussian white noise series with mean zero and variance 0.02. What are the mean and variance of the return series $r_t$? Compute the lag-1 and lag-2 autocorrelations of $r_t$. Assume that $r_{100} = -0.01$, and $r_{99} = 0.02$. Compute the 1- and 2-step-ahead forecasts of the return series at the forecast origin $t = 100$. What are the associated standard deviations of the forecast errors?

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**Solution.**

**Notation.**

- $r_t$: daily log return; $a_t \sim N(0, \sigma_a^2)$ i.i.d. shock, $\sigma_a^2 = 0.02$.
- $\phi_0 = 0.01$, $\phi_2 = 0.2$ (the lag-1 coefficient is $\phi_1 = 0$).
- $\mu = \mathbb{E}(r_t)$; $\gamma_\ell = \mathrm{Cov}(r_t, r_{t-\ell})$; $\rho_\ell = \gamma_\ell/\gamma_0$, $\ell \geq 0$.
- $\hat{r}_{100}(\ell) = \mathbb{E}(r_{100+\ell} \mid r_{100}, r_{99}, \dots)$: $\ell$-step forecast at origin 100; $e_{100}(\ell) = r_{100+\ell} - \hat{r}_{100}(\ell)$: forecast error.

**Step — Model and stationarity.** The model is

$$r_t = \phi_0 + \phi_2 r_{t-2} + a_t. \tag{1.1}$$

Its characteristic equation $1 - \phi_2 x^2 = 0$ has roots $x = \pm 1/\sqrt{\phi_2} = \pm 1/\sqrt{0.2} = \pm 2.2361$, both outside the unit circle, so $r_t$ is weakly stationary and $\mu$, $\gamma_\ell$ do not depend on $t$.

**Step — Mean.** Taking expectations of (1.1) and using $\mathbb{E}(a_t) = 0$:

$$\mu = \phi_0 + \phi_2 \mu. \tag{1.2}$$

Solving (1.2) for $\mu$:

$$\mu = \frac{\phi_0}{1 - \phi_2}. \tag{1.3}$$

Substituting the numbers into (1.3):

$$\mu = \frac{(0.01)}{1 - (0.2)} = \frac{0.01}{0.8} = 0.0125.$$

**Step — Autocovariance equations.** Subtracting (1.2) from (1.1):

$$r_t - \mu = \phi_2 (r_{t-2} - \mu) + a_t. \tag{1.4}$$

Multiplying (1.4) by $(r_{t-\ell} - \mu)$ and taking expectations:

$$\gamma_\ell = \phi_2\,\gamma_{\ell-2} + \mathbb{E}\big[a_t(r_{t-\ell} - \mu)\big], \qquad \mathbb{E}\big[a_t(r_{t-\ell} - \mu)\big] = \begin{cases} \sigma_a^2, & \ell = 0, \\ 0, & \ell \geq 1, \end{cases} \tag{1.5}$$

because $a_t$ is uncorrelated with the past and $\mathbb{E}[a_t(r_t - \mu)] = \mathbb{E}(a_t^2) = \sigma_a^2$ by (1.4).
**Step — Variance.** Setting $\ell = 0$ and $\ell = 2$ in (1.5), with $\gamma_{-2} = \gamma_2$:

$$\gamma_0 = \phi_2\gamma_2 + \sigma_a^2, \tag{1.6}$$
$$\gamma_2 = \phi_2\gamma_0. \tag{1.7}$$

Substituting (1.7) into (1.6):

$$\gamma_0 = \phi_2\,(\phi_2\gamma_0) + \sigma_a^2.$$

Collecting $\gamma_0$:

$$\gamma_0 = \frac{\sigma_a^2}{1 - \phi_2^2}. \tag{1.8}$$

Substituting the numbers into (1.8):

$$\gamma_0 = \frac{(0.02)}{1 - (0.2)^2} = \frac{0.02}{0.96} = 0.020833, \qquad \sqrt{\gamma_0} = 0.1443.$$

**Step — Autocorrelations.** Setting $\ell = 1$ in (1.5), with $\gamma_{-1} = \gamma_1$:

$$\gamma_1 = \phi_2\gamma_1 \implies (1 - \phi_2)\gamma_1 = 0 \implies \gamma_1 = 0, \tag{1.9}$$

since $\phi_2 = 0.2 \neq 1$. Hence $\rho_1 = \gamma_1/\gamma_0 = 0$. Dividing (1.7) by $\gamma_0$:

$$\rho_2 = \frac{\gamma_2}{\gamma_0} = \phi_2 = 0.2.$$

**Step — 1-step forecast.** Writing (1.1) at $t = 101$:

$$r_{101} = \phi_0 + \phi_2 r_{99} + a_{101}. \tag{1.10}$$

$r_{99}$ is known at the origin and $\mathbb{E}(a_{101} \mid \cdot) = 0$, so

$$\hat{r}_{100}(1) = \phi_0 + \phi_2 r_{99}. \tag{1.11}$$

Substituting the numbers into (1.11):

$$\hat{r}_{100}(1) = 0.01 + 0.2\,(0.02) = 0.01 + 0.004 = 0.014.$$

**Step — 2-step forecast.** Writing (1.1) at $t = 102$:

$$r_{102} = \phi_0 + \phi_2 r_{100} + a_{102}, \tag{1.12}$$

and $r_{100}$ is known, so

$$\hat{r}_{100}(2) = \phi_0 + \phi_2 r_{100}. \tag{1.13}$$

Substituting the numbers into (1.13):

$$\hat{r}_{100}(2) = 0.01 + 0.2\,(-0.01) = 0.01 - 0.002 = 0.008.$$

**Step — Forecast-error standard deviations.** Subtracting (1.11) from (1.10) and (1.13) from (1.12):

$$e_{100}(1) = a_{101}, \qquad e_{100}(2) = a_{102}. \tag{1.14}$$

Both have variance $\sigma_a^2$, so

$$\mathrm{sd}[e_{100}(1)] = \mathrm{sd}[e_{100}(2)] = \sqrt{0.02} = 0.1414.$$

(In a general AR(2), $e_h(2) = a_{h+2} + \phi_1 a_{h+1}$; here $\phi_1 = 0$, so the 2-step error is as small as the 1-step error.)

*Intuition.* With $\phi_1 = 0$ the series is two interleaved AR(1) processes, one on odd and one on even days. Returns two days apart are correlated ($\rho_2 = 0.2$), adjacent days are not ($\rho_1 = 0$). The 2-step forecast is driven by a return already observed ($r_{100}$), so it is as precise as the 1-step one; both forecasts sit close to $\mu = 0.0125$ because $\phi_2$ is small, and the error standard deviation 0.1414 is only slightly below the unconditional 0.1443.

**Ref:** Tsay (2010), §2.4.1 (properties of AR models, pp. 38–46) and §2.4.4 (forecasting, pp. 54–56).

## Block 2 — Wooldridge, *Introductory Econometrics* *(W 11.C8; W 12.1)*

**Question 2.** W 11.C8 — AR(1) for unemployment (computer exercise).
**Source:** Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Cengage, Chapter 11, Computer Exercise C8, p. 370 — reproduced verbatim.

C8 Use the data in PHILLIPS for this exercise.

(i) Estimate an AR(1) model for the unemployment rate. Use this equation to predict the unemployment rate for 2004. Compare this with the actual unemployment rate for 2004. (You can find this information in a recent *Economic Report of the President*.)

(ii) Add a lag of inflation to the AR(1) model from part (i). Is $inf_{t-1}$ statistically significant?

(iii) Use the equation from part (ii) to predict the unemployment rate for 2004. Is the result better or worse than in the model from part (i)?

(iv) Use the method from Section 6-4 to construct a 95% prediction interval for the 2004 unemployment rate. Is the 2004 unemployment rate in the interval?

**Solution.**

**Notation.**

- $unem_t$: U.S. civilian unemployment rate in year $t$ (percent); $inf_t$: CPI inflation rate in year $t$ (percent). PHILLIPS covers 1948–2003; with one lag the estimation sample is 1949–2003, $n = 55$.
- $\beta_j$: population coefficients; $\hat\beta_j$: OLS estimates; $\mathrm{se}(\cdot)$: usual OLS standard error; $u_t$: error; $\hat\sigma^2 = \mathrm{SSR}/(n-k-1)$; $R^2$.
- $\hat y_0$: prediction of $unem_{2004}$; $\hat e_0 = unem_{2004} - \hat y_0$: prediction error; $\theta_0$: population value of the prediction. Actual 2004 rate: 5.5 (BLS annual average, as reported in the *Economic Report of the President*; the mean of the 12 monthly FRED `UNRATE` values for 2004 is 5.54).
- 2003 values used for the 2004 prediction: $unem_{2003} = 6.0$, $inf_{2003} = 2.3$.

**Step — (i) AR(1) estimates.** The model is

$$unem_t = \beta_0 + \beta_1\, unem_{t-1} + u_t. \tag{2.1}$$

OLS on 1949–2003 gives

$$\widehat{unem}_t = \underset{(0.5202)}{1.489685} + \underset{(0.0893)}{0.742382}\, unem_{t-1}, \qquad n = 55, \quad R^2 = 0.5660, \quad \hat\sigma = 0.9986. \tag{2.2}$$

**Step — (i) Prediction for 2004.** Evaluating (2.2) at $unem_{2003}$:

$$\widehat{unem}_{2004} = \hat\beta_0 + \hat\beta_1\, unem_{2003}. \tag{2.3}$$

Substituting the numbers into (2.3):

$$\widehat{unem}_{2004} = 1.489685 + 0.742382\,(6.0) = 1.489685 + 4.454292 = 5.9440.$$

Prediction error against the actual rate:

$$\hat{y}_0 - unem_{2004} = 5.9440 - 5.5 = 0.4440,$$

so the AR(1) overpredicts 2004 unemployment by about 0.44 percentage points.

**Step — (ii) Adding lagged inflation.** The model is

$$unem_t = \beta_0 + \beta_1\, unem_{t-1} + \beta_2\, inf_{t-1} + u_t, \tag{2.4}$$

with OLS estimates

$$\widehat{unem}_t = \underset{(0.4413)}{1.296416} + \underset{(0.0779)}{0.649479}\, unem_{t-1} + \underset{(0.0388)}{0.183009}\, inf_{t-1}, \qquad n = 55, \quad R^2 = 0.6963. \tag{2.5}$$

The $t$ statistic for $H_0 : \beta_2 = 0$ is

$$t_{\hat{\beta}_2} = \frac{\hat{\beta}_2}{\mathrm{se}(\hat{\beta}_2)} = \frac{0.183009}{0.0388} = 4.72, \tag{2.6}$$

against a two-sided 5% critical value $t_{52,\,0.975} = 2.0066$ (two-sided $p$-value 0.00002). So $inf_{t-1}$ is statistically significant at any conventional level. The positive sign says that a year of higher inflation is followed by higher unemployment, the pattern of an expectations-augmented Phillips curve with supply shocks, not a stable inflation–unemployment trade-off.

**Step — (iii) Prediction from** (2.5)**.** Evaluating (2.5) at the 2003 values:

$$\widehat{unem}_{2004} = \hat{\beta}_0 + \hat{\beta}_1\, unem_{2003} + \hat{\beta}_2\, inf_{2003}. \tag{2.7}$$

Substituting the numbers into (2.7):

$$\widehat{unem}_{2004} = 1.296416 + 0.649479\,(6.0) + 0.183009\,(2.3) = 1.296416 + 3.896874 + 0.420921 = 5.6142.$$

Prediction error:

$$\hat{y}_0 - unem_{2004} = 5.6142 - 5.5 = 0.1142.$$

Since $0.1142 < 0.4440$, the model with lagged inflation predicts 2004 better (for this one year).

**Step — (iv) Section 6-4: the prediction and its standard error.** The population prediction is

$$\theta_0 = \beta_0 + \beta_1(6.0) + \beta_2(2.3). \tag{2.8}$$

Solving (2.8) for $\beta_0$ and substituting into (2.4):

$$unem_t = \theta_0 + \beta_1\,(unem_{t-1} - 6.0) + \beta_2\,(inf_{t-1} - 2.3) + u_t, \tag{2.9}$$

so the OLS intercept of (2.9) is $\hat{y}_0 = 5.6142$ (the same number as in (iii)) and its standard error is $\mathrm{se}(\hat{y}_0) = 0.136381$.

**Step — (iv) Standard error of the prediction error.** The prediction error $\hat{e}_0 = u_{2004} - (\hat{y}_0 - \theta_0)$ adds the new error to the estimation error, which are uncorrelated, so (Wooldridge eq. (6.36))

$$\mathrm{se}(\hat{e}_0) = \sqrt{\mathrm{se}(\hat{y}_0)^2 + \hat{\sigma}^2}. \tag{2.10}$$

With $\hat{\sigma}^2 = \mathrm{SSR}/52 = 0.711321$, substituting into (2.10):

$$\mathrm{se}(\hat{e}_0) = \sqrt{(0.136381)^2 + (0.711321)} = \sqrt{0.018600 + 0.711321} = \sqrt{0.729921} = 0.8544.$$

**Step — (iv) 95% prediction interval.** The interval (Wooldridge eq. (6.37)) is

$$\hat{y}_0 \pm t_{52,\,0.975} \cdot \mathrm{se}(\hat{e}_0). \tag{2.11}$$

Substituting into (2.11): the half-width is $2.0066\,(0.8544) = 1.7144$, so

$$[\,5.6142 - 1.7144,\ 5.6142 + 1.7144\,] = [\,3.8998,\ 7.3286\,].$$

The actual 2004 rate, 5.5, lies inside the interval.

**Step — Python.** Code and key output (the full version, with assertions, is in the check script):

```
import wooldridge, statsmodels.api as sm, numpy as np
ph = wooldridge.data("phillips").sort_values("year")
ph["unem_1"], ph["inf_1"] = ph.unem.shift(1), ph.inf.shift(1)
e = ph.dropna(subset=["unem_1", "inf_1"])                    # 1949-2003, n = 55
m1 = sm.OLS(e.unem, sm.add_constant(e[["unem_1"]])).fit()    # 1.489685, 0.742382
m2 = sm.OLS(e.unem, sm.add_constant(e[["unem_1", "inf_1"]])).fit()
#   const 1.296416 (0.4413)  unem_1 0.649479 (0.0779)  inf_1 0.183009 (0.0388), t = 4.72
X = np.column_stack([np.ones(55), e.unem_1 - 6.0, e.inf_1 - 2.3])
m3 = sm.OLS(e.unem.values, X).fit()                          # 5.6142, se 0.136381
se_e0 = np.sqrt(m3.bse[0]**2 + m3.scale)        # 0.8544 -> PI [3.8998, 7.3286]
```

*Intuition.* Unemployment is persistent ($\hat{\beta}_1 \approx 0.74$), so an AR(1) mostly extrapolates 2003's 6.0%. Lagged inflation carries extra information about where the cycle is heading, which lowers the 2004 forecast toward the realised 5.5%. The prediction interval is wide because its width is dominated by $\hat{\sigma}^2 = 0.711$, the variance of next year's shock; parameter uncertainty (0.0186) is a small part of it. One year of forecast comparison is weak evidence: a fair comparison would use many out-of-sample years.

**Ref:** Wooldridge (2016), §11-2 and Example 11.5 (dynamic models, PHILLIPS), §18-5 (forecasting), §6-4a (prediction intervals, pp. 186–190).

**Question 3.** W 12.1 — OLS standard errors under AR(1) errors.
**Source:** Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Cengage, Chapter 12, Problem 1, p. 396 — reproduced verbatim.

1 When the errors in a regression model have AR(1) serial correlation, why do the OLS standard errors tend to underestimate the sampling variation in the $\hat{\beta}_j$? Is it always true that the OLS standard errors are too small?

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**Solution.**

**Notation.**

- $y_t = \beta_0 + \beta_1 x_t + u_t$, $t = 1, \ldots, n$; $x_t$ treated as nonrandom (condition on $X$), with $\bar{x} = 0$ for simplicity (as in Wooldridge §12-1b).
- $u_t = \rho u_{t-1} + e_t$, $|\rho| < 1$, $e_t$ i.i.d. $(0, \sigma_e^2)$; $\sigma^2 = \mathrm{Var}(u_t) = \sigma_e^2/(1-\rho^2)$.
- $\mathrm{SST}_x = \sum_t x_t^2$; $\lambda$: first-order autocorrelation of $x_t$ when $x_t$ is itself AR(1) (used only for the illustration).

**Step — Sampling error of the slope.** With $\bar{x} = 0$, OLS gives

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{t=1}^{n} x_t u_t}{\mathrm{SST}_x}. \tag{3.1}$$

Under the AR(1) model the error covariances are

$$\mathrm{Cov}(u_t, u_{t+j}) = \rho^j \sigma^2, \qquad j \geq 0. \tag{3.2}$$

**Step — True variance.** Taking the variance of (3.1):

$$\mathrm{Var}(\hat{\beta}_1) = \frac{1}{\mathrm{SST}_x^2}\Big[\sum_t x_t^2\,\mathrm{Var}(u_t) + 2\sum_{t<s} x_t x_s\,\mathrm{Cov}(u_t, u_s)\Big]. \tag{3.3}$$

Substituting (3.2) into (3.3):

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\mathrm{SST}_x} + \frac{2\sigma^2}{\mathrm{SST}_x^2} \sum_{t=1}^{n-1} \sum_{j=1}^{n-t} \rho^j x_t x_{t+j}, \tag{3.4}$$

which is Wooldridge's equation (12.4).

**Step — What the usual formula estimates.** The usual OLS variance is

$$\widehat{\mathrm{Var}}_{\mathrm{OLS}}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{\mathrm{SST}_x}, \tag{3.5}$$

an estimate of the first term of (3.4) only. The usual standard error is therefore biased by the omitted second term.

**Step — Why it is usually too small.** In economic data $\rho > 0$ and the regressors are positively autocorrelated (persistent, trending), so $x_t x_{t+j} > 0$ on average and every $\rho^j > 0$: the double sum in (3.4) is positive and (3.5) understates the true variance. A large-$n$ illustration: if $x_t$ is AR(1) with autocorrelation $\lambda$, then $\sum_t \sum_j \rho^j x_t x_{t+j} \approx \mathrm{SST}_x \sum_{j \geq 1} (\rho\lambda)^j$, and substituting into (3.4):

$$\mathrm{Var}(\hat{\beta}_1) \approx \frac{\sigma^2}{\mathrm{SST}_x}\Big[1 + \frac{2\rho\lambda}{1-\rho\lambda}\Big] = \frac{\sigma^2}{\mathrm{SST}_x} \cdot \frac{1+\rho\lambda}{1-\rho\lambda}. \tag{3.6}$$

With $\rho = \lambda = 0.5$, (3.6) gives the ratio $(1+0.25)/(1-0.25) = 1.25/0.75 = 1.6667$: the true standard error is $\sqrt{1.6667} = 1.2910$ times the usual one.

**Step — Not always.** The sign of the second term in (3.4) is the sign of $\sum \rho^j x_t x_{t+j}$. If $\rho < 0$ (with persistent $x$), or if $\rho > 0$ but $x_t$ alternates in sign (negatively autocorrelated), the cross terms are negative on net and the usual standard error is too large. In (3.6), $\rho = -0.5$, $\lambda = 0.5$ gives $(1-0.25)/(1+0.25) = 0.75/1.25 = 0.6000$, so the true standard error is only $\sqrt{0.6} = 0.7746$ times the usual one. If $x_t$ is serially uncorrelated ($\lambda = 0$) the bias vanishes in large samples even with $\rho \neq 0$. So the answer to the second question is no.

*Intuition.* Positively correlated errors make neighbouring observations repeat each other, so the sample carries less independent information than $n$ suggests; the usual formula counts every observation as new. When regressor and error both drift slowly, a run of positive errors coincides with a run of large $x$, which tilts the fitted slope more than independent errors would.

**Ref:** Wooldridge (2016), §12-1b (efficiency and inference, pp. 373–374).

## Block 3 — Greenberg, *Introduction to Bayesian Econometrics* *(G 2.2; G 7.3)*

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**Question 4.** G 2.2 — gamma–Poisson conjugacy.

**Source:** Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, §2.5 Exercises, Exercise 2.2, p. 19 — reproduced verbatim.

2.2 The Poisson distribution has probability mass function

$$p(y_i \mid \theta) = \frac{\theta^{y_i} e^{-\theta}}{y_i!}, \qquad \theta > 0,\ y_i = 0, 1, \ldots,$$

and let $y_1, \ldots, y_n$ be a random sample from this distribution.

**(a)** Show that the gamma distribution $G(\alpha, \beta)$ is a conjugate prior distribution for the Poisson distribution.

**(b)** Show that $\bar{y}$ is the MLE for $\theta$.

**(c)** Write the mean of the posterior distribution as a weighted average of the mean of the prior distribution and the MLE.

**(d)** What happens to the weight on the prior mean as $n$ becomes large?

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**Solution.**

**Notation.**

- $y_i \in \{0, 1, \dots\}$: counts; $\theta > 0$: Poisson rate; $s = \sum_{i=1}^{n} y_i$, $\bar{y} = s/n$.
- $G(\alpha, \beta)$, $\alpha, \beta > 0$: gamma with shape $\alpha$ and rate $\beta$ (Greenberg's convention, App. A), $\pi(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\theta^{\alpha-1}e^{-\beta\theta}$, mean $\alpha/\beta$.
- $\ell(\theta)$: log-likelihood; $\hat{\theta}$: MLE; $w$: weight on the prior mean.

**Step — (a) Likelihood.** Independence gives

$$p(y \mid \theta) = \prod_{i=1}^{n} \frac{\theta^{y_i}e^{-\theta}}{y_i!} = \frac{\theta^s e^{-n\theta}}{\prod_i y_i!}. \tag{4.1}$$

**Step — (a) Posterior.** Bayes' theorem, $\pi(\theta \mid y) \propto p(y \mid \theta)\pi(\theta)$, with (4.1) and the gamma prior, dropping factors free of $\theta$:

$$\pi(\theta \mid y) \propto \left(\theta^s e^{-n\theta}\right)\left(\theta^{\alpha-1}e^{-\beta\theta}\right). \tag{4.2}$$

Adding exponents in (4.2):

$$\pi(\theta \mid y) \propto \theta^{\alpha+s-1}e^{-(\beta+n)\theta}, \tag{4.3}$$

the kernel of $G(\alpha+s,\ \beta+n)$. Prior and posterior are both gamma, so the gamma family is conjugate for the Poisson likelihood.

**Step — (b) MLE.** Taking logs of (4.1):

$$\ell(\theta) = s\log\theta - n\theta - \sum_i \log y_i!. \tag{4.4}$$

Differentiating (4.4) and setting the score to zero:

$$\ell'(\theta) = \frac{s}{\theta} - n = 0 \implies \hat{\theta} = \frac{s}{n} = \bar{y}. \tag{4.5}$$

The second derivative $\ell''(\theta) = -s/\theta^2 < 0$ when $s > 0$, so (4.5) is a maximum. (If $s = 0$, $\ell(\theta) = -n\theta$ decreases and the supremum is at the boundary $0 = \bar{y}$.)

**Step — (c) Posterior mean as a weighted average.** The mean of the gamma (4.3) is

$$\mathbb{E}(\theta \mid y) = \frac{\alpha + s}{\beta + n}. \tag{4.6}$$

Substituting $\alpha = \beta(\alpha/\beta)$ and $s = n\bar{y}$ into (4.6):

$$\mathbb{E}(\theta \mid y) = \frac{\beta\,(\alpha/\beta) + n\,(\bar{y})}{\beta + n}.$$

Splitting the fraction:

$$\mathbb{E}(\theta \mid y) = \underbrace{\frac{\beta}{\beta+n}}_{w} \cdot \frac{\alpha}{\beta} + \underbrace{\frac{n}{\beta+n}}_{1-w} \cdot \bar{y}. \tag{4.7}$$

**Step — (d) Large $n$.** From (4.7),

$$w = \frac{\beta}{\beta+n} \xrightarrow[n\to\infty]{} 0, \tag{4.8}$$

so the posterior mean converges to the MLE $\bar{y}$; the posterior variance $(\alpha+s)/(\beta+n)^2$ also goes to zero at rate $1/n$.

**Step — Numerical illustration.** Coal-mining disasters 1851–1891 ($n = 41$, $s = 127$, $\bar{y} = 3.097561$) with the prior of G 7.3, $\alpha = 0.5$, $\beta = 1$. By (4.8), $w = 1/(1 + 41) = 0.023810$; by (4.7),

$$\mathbb{E}(\theta \mid y) = 0.023810\,(0.5) + 0.976190\,(3.097561) = 0.011905 + 3.023810 = 3.035714,$$

which equals (4.6): $(0.5 + 127)/(1 + 41) = 127.5/42 = 3.035714$.

*Intuition.* The prior acts like $\beta$ extra periods of data containing $\alpha$ events in total. After $n$ real periods it weighs $\beta/(\beta + n)$; with 41 years of data a prior worth one year moves the estimate only from 3.098 to 3.036.

**Ref:** Greenberg (2008), §2.2 (pp. 12–18) and App. A (gamma distribution); the Class 1 deck *first Bayesian example* (Poisson with an exponential prior, the case $\alpha = 1$).

**Question 5.** G 7.3 — Gibbs sampler for the coal-mining switch point (computer exercise).
**Source:** Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, §7.5 Exercises, Exercise 7.3, p. 106 — reproduced verbatim. Equations (7.3) and (7.4), to which the exercise refers, are reproduced verbatim from §7.1.1, pp. 93–94.

*Equations referred to (Greenberg 2008, pp. 93–94):*

$$p(y_i) = \begin{cases} \dfrac{e^{-\theta_1}\theta_1^{y_i}}{y_i!}, & \text{for } i = 1, \ldots, k, \\ \dfrac{e^{-\theta_2}\theta_2^{y_i}}{y_i!}, & \text{for } i = k+1, \ldots, n, \end{cases} \tag{7.3}$$

"where $y_i = 0, 1, \ldots,$ and the switch point $k$ is unknown. The specification is completed by assigning the conditionally conjugate priors,

$$\theta_1 \sim G(\alpha_{10}, \beta_{10}), \quad \theta_2 \sim G(\alpha_{20}, \beta_{20}), \quad \pi(k = j) = 1/n, \quad j = 1, \ldots, n." \tag{7.4}$$

7.3 Construct a Gibbs algorithm to analyze the Poisson model with unknown switch point. Given the specification in Equations (7.3) and (7.4), show that

$$\pi(\theta_1, \theta_2, k \mid y) \propto \theta_1^{\alpha_{10}-1} e^{-\beta_{10}\theta_1} \theta_2^{\alpha_{20}-1} e^{-\beta_{20}\theta_2} \prod_{i=1}^{k} e^{-\theta_1}\theta_1^{y_i} \prod_{i=k+1}^{n} e^{-\theta_2}\theta_2^{y_i},$$

and verify that

$$\theta_1 \mid y, k \sim G\Big(\alpha_{10} + \sum_{1}^{k} y_i,\ \beta_{10} + k\Big),$$

$$\theta_2 \mid k, y \sim G\Big(\alpha_{20} + \sum_{k+1}^{n} y_i,\ \beta_{20} + n - k\Big),$$

$$\pi(k \mid y, \theta_1, \theta_2) = \frac{e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{\sum_1^k y_i}}{\sum_{k=1}^{n} e^{k(\theta_2-\theta_1)}\,(\theta_1/\theta_2)^{\sum_1^k y_i}}, \qquad k = 1, \ldots, n.$$

Apply your algorithm to the mining disaster data analyzed by Carlin et al. (1992a); the data may be found in the article. Your analysis should generate marginal posterior distributions for $\theta_1$, $\theta_2$, $p(k)$, and the marginal likelihood computed by the method of Chib (1995). For simplicity, set $\alpha_{10} = \alpha_{20} = 0.5$ and $\beta_{10} = \beta_{20} = 1.0$. (Curiously, although these data have been analyzed from the Bayesian viewpoint in several papers, none of them present historical material that might shed light on when changes in parameters took place. For example, changes in government regulation, technology, or the location of mines might be expected to affect the probability of a disaster. A more general version of the model permits multiple switch points if necessary.)

---

**Solution.**

**Notation.**

- $y_i$: number of British coal-mining disasters in year $i$, $i = 1$ (1851) to $n = 112$ (1962); the series of Carlin, Gelfand & Smith (1992) is Jarrett (1979), total $S = \sum_i y_i = 191$.
- $k \in \{1, \dots, n\}$: switch point, the last year of regime 1 (year $1850 + k$); $k = n$ means no change.
- $\theta_1, \theta_2 > 0$: disaster rates before and after the switch; $S_k = \sum_{i=1}^{k} y_i$ (so $\sum_{k+1}^{n} y_i = S - S_k$).
- $G(a, b)$: gamma with shape $a$, rate $b$; $\alpha_{10} = \alpha_{20} = \alpha = 0.5$, $\beta_{10} = \beta_{20} = \beta = 1$.
- $g = 1, \dots, G$: Gibbs iteration; $\theta^* = (\theta_1^*, \theta_2^*)$: point where Chib's identity is evaluated; $m(y)$: marginal likelihood.

**Step — Joint posterior.** By (7.3) and independence, the likelihood is

$$p(y \mid \theta_1, \theta_2, k) = \prod_{i=1}^{k} \frac{e^{-\theta_1}\theta_1^{y_i}}{y_i!} \prod_{i=k+1}^{n} \frac{e^{-\theta_2}\theta_2^{y_i}}{y_i!}, \tag{5.1}$$

and by (7.4) the prior is

$$\pi(\theta_1, \theta_2, k) = \frac{\beta_{10}^{\alpha_{10}}}{\Gamma(\alpha_{10})}\theta_1^{\alpha_{10}-1}e^{-\beta_{10}\theta_1} \cdot \frac{\beta_{20}^{\alpha_{20}}}{\Gamma(\alpha_{20})}\theta_2^{\alpha_{20}-1}e^{-\beta_{20}\theta_2} \cdot \frac{1}{n}. \tag{5.2}$$

Multiplying (5.1) by (5.2) and dropping the factors free of $(\theta_1, \theta_2, k)$ (the $y_i!$, the gamma constants, $1/n$) gives the expression in the statement:

$$\pi(\theta_1, \theta_2, k \mid y) \propto \theta_1^{\alpha_{10}-1}e^{-\beta_{10}\theta_1}\theta_2^{\alpha_{20}-1}e^{-\beta_{20}\theta_2} \prod_{i=1}^{k} e^{-\theta_1}\theta_1^{y_i} \prod_{i=k+1}^{n} e^{-\theta_2}\theta_2^{y_i}. \tag{5.3}$$

**Step — Full conditional of $\theta_1$.** Keeping in (5.3) only the factors with $\theta_1$:

$$\pi(\theta_1 \mid y, k, \theta_2) \propto \theta_1^{\alpha_{10}-1}e^{-\beta_{10}\theta_1}\big(e^{-k\theta_1}\theta_1^{S_k}\big).$$

Adding exponents:

$$\pi(\theta_1 \mid y, k) \propto \theta_1^{\alpha_{10}+S_k-1}e^{-(\beta_{10}+k)\theta_1} \implies \theta_1 \mid y, k \sim G(\alpha_{10} + S_k,\ \beta_{10} + k). \tag{5.4}$$

**Step — Full conditional of $\theta_2$.** In the same way, the $\theta_2$ factors of (5.3) are

$$\pi(\theta_2 \mid y, k, \theta_1) \propto \theta_2^{\alpha_{20}-1}e^{-\beta_{20}\theta_2}\big(e^{-(n-k)\theta_2}\theta_2^{S-S_k}\big),$$

so, adding exponents,

$$\theta_2 \mid y, k \sim G(\alpha_{20} + S - S_k,\ \beta_{20} + n - k). \tag{5.5}$$

Neither (5.4) nor (5.5) involves the other rate: given $k$, $\theta_1$ and $\theta_2$ are independent.

**Step — Full conditional of $k$.** The factors of (5.3) that change with $k$ are the two products:

$$\pi(k \mid y, \theta_1, \theta_2) \propto e^{-k\theta_1}\theta_1^{S_k}\, e^{-(n-k)\theta_2}\theta_2^{S-S_k}. \tag{5.6}$$

Dividing (5.6) by the $k$-free factor $e^{-n\theta_2}\theta_2^{S}$:

$$\pi(k \mid y, \theta_1, \theta_2) \propto e^{k(\theta_2-\theta_1)}\Big(\frac{\theta_1}{\theta_2}\Big)^{S_k}, \qquad k = 1, \dots, n, \tag{5.7}$$

and normalising over $k = 1, \dots, n$ gives the formula in the statement.

**Step — Algorithm.** Start at $k^{(0)} = 56$. For $g = 1, \dots, 11{,}000$: draw $\theta_1^{(g)}$ from (5.4) and $\theta_2^{(g)}$ from (5.5) given $k^{(g-1)}$; compute the $n$ weights (5.7) on the log scale, normalise, and draw $k^{(g)}$. Discard 1,000 burn-in draws and keep $G = 10{,}000$ (NumPy seed 2026).

**Step — Exact benchmark.** Because the priors are conjugate given $k$, $\theta_1, \theta_2$ integrate out of (5.1)×(5.2) in closed form:

$$p(y \mid k) = \frac{1}{\prod_i y_i!} \cdot \frac{\beta^{\alpha}\,\Gamma(\alpha + S_k)}{\Gamma(\alpha)\,(\beta + k)^{\alpha+S_k}} \cdot \frac{\beta^{\alpha}\,\Gamma(\alpha + S - S_k)}{\Gamma(\alpha)\,(\beta + n - k)^{\alpha+S-S_k}}, \tag{5.8}$$

so the exact marginal likelihood and posterior of $k$ are

$$m(y) = \sum_{k=1}^{n} \frac{1}{n}\, p(y \mid k), \qquad p(k \mid y) = \frac{p(y \mid k)/n}{m(y)}, \tag{5.9}$$

giving $\log m(y) = -179.4590$, $\mathbb{E}(\theta_1 \mid y) = 3.0500$, $\mathbb{E}(\theta_2 \mid y) = 0.9147$. These check the sampler.

**Step — Marginal posteriors.** Gibbs output (exact values from (5.9) for $k$):

| | Posterior mean | Posterior sd | 95% interval |
|---|---|---|---|
| $\theta_1$ (before) | 3.0506 | 0.2809 | [2.530, 3.635] |
| $\theta_2$ (after) | 0.9166 | 0.1163 | [0.701, 1.156] |

| $k$ | year | $p(k \mid y)$ Gibbs | exact |
|---|---|---|---|
| 41 | 1891 | 0.2471 | 0.2482 |
| 40 | 1890 | 0.1849 | 0.1848 |
| 39 | 1889 | 0.1433 | 0.1414 |
| 42 | 1892 | 0.0987 | 0.1010 |
| 37 | 1887 | 0.0881 | 0.0898 |
| 36 | 1886 | 0.0742 | 0.0745 |

The posterior of $k$ has mean 40.14 and mode $k = 41$ (1891); $P(36 \le k \le 46 \mid y) = 0.98$, i.e. the switch happened between 1886 and 1896 with high probability. Disasters fell from about 3.05 to about 0.92 per year.

**Step — Marginal likelihood by Chib (1995).** For any $\theta^*$, Bayes' theorem rearranged gives the identity

$$\log m(y) = \log p(y \mid \theta^*) + \log \pi(\theta^*) - \log \pi(\theta^* \mid y). \tag{5.10}$$

Here $k$ is a latent parameter, so the likelihood term sums it out with its prior:

$$p(y \mid \theta^*) = \sum_{k=1}^{n} \frac{1}{n}\, p(y \mid \theta_1^*, \theta_2^*, k). \tag{5.11}$$

For the ordinate, conditional independence of $\theta_1, \theta_2$ given $k$ (from (5.4) and (5.5)) gives the Rao–Blackwell estimate

$$\hat{\pi}(\theta^* \mid y) = \frac{1}{G} \sum_{g=1}^{G} G\big(\theta_1^* \mid \alpha + S_{k^{(g)}}, \beta + k^{(g)}\big)\, G\big(\theta_2^* \mid \alpha + S - S_{k^{(g)}}, \beta + n - k^{(g)}\big), \tag{5.12}$$

with $G(\cdot \mid a, b)$ the gamma density. At the posterior means $\theta^* = (3.0506,\ 0.9166)$: $\log p(y \mid \theta^*) = -172.2563$ from (5.11), $\log \pi(\theta^*) = -5.6261$ from (7.4), and $\log \hat{\pi}(\theta^* \mid y) = 1.5765$ from (5.12). Substituting into (5.10):

$$\log \hat{m}(y) = (-172.2563) + (-5.6261) - (1.5765) = -179.4589,$$

against the exact $-179.4590$ from (5.9).

**Step — Python.** Core of the sampler (complete code with assertions in the check script):

```
cs, ks = np.cumsum(y), np.arange(1, n + 1)        # S_k and k = 1..n
rng, k = np.random.default_rng(2026), n // 2
for g in range(11000):
    t1 = rng.gamma(a0 + cs[k-1], 1 / (b0 + k))                # (5.4) theta1 | k
    t2 = rng.gamma(a0 + S - cs[k-1], 1 / (b0 + n - k))        # (5.5) theta2 | k
    lp = ks * (t2 - t1) + cs * np.log(t1 / t2)                # (5.7) log pi(k | ...)
    p = np.exp(lp - lp.max()); k = rng.choice(ks, p=p / p.sum())
# output: theta1 3.0506 (0.2809), theta2 0.9166 (0.1163), mode k = 41 (1891)
# Chib: -172.2563 + (-5.6261) - 1.5765 = -179.4589 ; exact -179.4590
```

*Intuition.* The data choose a sharp break: about three disasters a year until around 1890, under one afterwards, with the switch year pinned down within a few years. The Gibbs sampler works because each block is easy given the others: rates are conjugate gammas given $k$, and $k$ is a discrete draw over 112 years given the rates. On the historical question Greenberg raises, the posterior mass near

1887–1891 sits close to the Coal Mines Regulation Act of 1887, a candidate explanation that this model does not test. Chib's identity recovers the exact marginal likelihood to the fourth decimal, which is what makes it usable for comparing this model with, say, a no-break Poisson in Classes 3+4.

**Ref:** Greenberg (2008), §7.1.1 (pp. 91–95) and §7.4 (Chib 1995); Class 1 deck *Poisson with breakpoint* and `Notas/L01/L01-03-coal-mining-change-point` (the profile likelihood also peaks at 1891).

## Block 4 — James, Witten, Hastie, Tibshirani & Taylor, *ISLP* *(I 5.2; I 5.9)*

**Question 6.** I 5.2 — probability that an observation is in a bootstrap sample.
**Source:** James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. (2023), *An Introduction to Statistical Learning with Applications in Python*, Springer, §5.4 Exercises (Conceptual), Exercise 2, pp. 224–225 — reproduced verbatim (the margin index entry "`np.empty()`" is omitted).

2. We will now derive the probability that a given observation is part of a bootstrap sample. Suppose that we obtain a bootstrap sample from a set of $n$ observations.

(a) What is the probability that the first bootstrap observation is *not* the $j$th observation from the original sample? Justify your answer.

(b) What is the probability that the second bootstrap observation is *not* the $j$th observation from the original sample?

(c) Argue that the probability that the $j$th observation is *not* in the bootstrap sample is $(1-1/n)^n$.

(d) When $n = 5$, what is the probability that the $j$th observation is in the bootstrap sample?

(e) When $n = 100$, what is the probability that the $j$th observation is in the bootstrap sample?

(f) When $n = 10{,}000$, what is the probability that the $j$th observation is in the bootstrap sample?

(g) Create a plot that displays, for each integer value of $n$ from 1 to 100,000, the probability that the $j$th observation is in the bootstrap sample. Comment on what you observe.

(h) We will now investigate numerically the probability that a bootstrap sample of size $n = 100$ contains the $j$th observation. Here $j = 4$. We first create an array `store` with values that will subsequently be overwritten using the function `np.empty()`. We then repeatedly create bootstrap samples, and each time we record whether or not the fifth observation is contained in the bootstrap sample.

```
rng = np.random.default_rng(10)
store = np.empty(10000)
for i in range(10000):
    store[i] = np.sum(rng.choice(100, replace=True) == 4)
          > 0
np.mean(store)
```

Comment on the results obtained.

**Solution.**

**Notation.**

- $n$: sample size; $j \in \{1, \dots, n\}$: a fixed original observation.
- $Z_1, \dots, Z_n$: indices of the bootstrap draws, i.i.d. uniform on $\{1, \dots, n\}$ (sampling with replacement).
- $p_n = P(j \text{ is in the bootstrap sample})$.

**Step — (a).** Each draw picks one of the $n$ original observations with equal probability $1/n$, so

$$P(Z_1 \neq j) = 1 - \frac{1}{n}. \tag{6.1}$$

**Step — (b).** Sampling is with replacement, so $Z_2$ has the same distribution as $Z_1$ and is independent of it:

$$P(Z_2 \neq j) = 1 - \frac{1}{n}. \tag{6.2}$$

**Step — (c).** $j$ is absent only if every draw misses it; by independence and (6.1)–(6.2) applied to all $n$ draws:

$$P(j \notin \text{sample}) = \prod_{m=1}^{n} P(Z_m \neq j) = \Big(1 - \frac{1}{n}\Big)^n. \tag{6.3}$$

**Step — (d)–(f).** Taking the complement of (6.3):

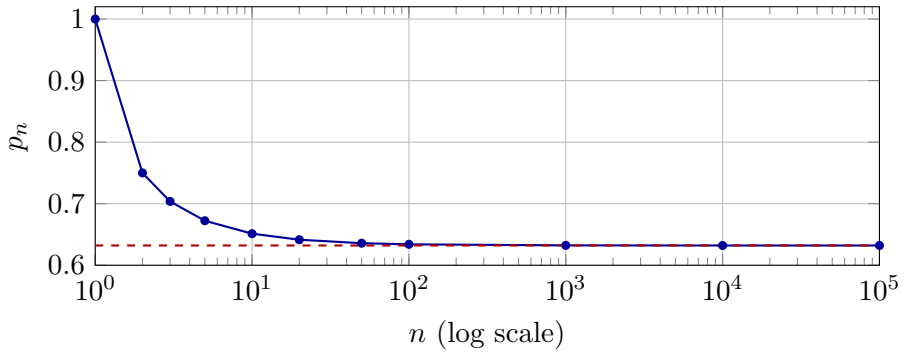
$$p_n = 1 - \Big(1 - \frac{1}{n}\Big)^n. \tag{6.4}$$

Substituting into (6.4):

$$\begin{aligned} n = 5: &\quad p_5 = 1 - (4/5)^5 = 1 - 0.32768 = 0.67232, \\ n = 100: &\quad p_{100} = 1 - (0.99)^{100} = 1 - 0.366032 = 0.633968, \\ n = 10{,}000: &\quad p_{10000} = 1 - (0.9999)^{10000} = 1 - 0.367861 = 0.632139. \end{aligned}$$

**Step — (g) Plot and limit.** Since $n\log(1-1/n) = n\big(-\frac{1}{n} - \frac{1}{2n^2} - \dots\big) \to -1$,

$$\lim_{n\to\infty} p_n = 1 - e^{-1} = 0.632121. \tag{6.5}$$




$p_n$ falls from 1 at $n = 1$ and flattens almost immediately at $1 - e^{-1} \approx 0.632$ (dashed): for $n \geq 100$ the curve is indistinguishable from the limit.

**Step — (h) The printed code.** `rng.choice(100, replace=True)` without `size` returns a *single* integer, so each iteration checks whether one draw equals 4, an event of probability 1/100. Run as printed, `np.mean(store)` returns 0.0089, not a bootstrap probability. The intended code draws a whole bootstrap sample, `rng.choice(100, size=100, replace=True)`, and then returns 0.6362, close to $p_{100} = 0.633968$ from (e) (Monte Carlo standard error $\sqrt{0.634 \cdot 0.366/10000} \approx 0.005$). Index 4 is the fifth observation because Python counts from 0, which is why the text says "fifth".

*Intuition.* A bootstrap sample of size $n$ contains only about 63.2% of the distinct original observations; the other 36.8% are "out of bag". This is why bootstrap estimates of test error that reuse the training points are optimistic, and why out-of-bag points can serve as a free validation set.

**Ref:** ISLP (2023), §5.2 (the bootstrap, pp. 212–215).

**Question 7.** I 5.9 — bootstrap standard errors for Boston `medv` (computer exercise).
**Source:** James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. (2023), *An Introduction to Statistical Learning with Applications in Python*, Springer, §5.4 Exercises (Applied), Exercise 9, pp. 227–228 — reproduced verbatim (the margin index entry "`np.percentile()`" is omitted).

9. We will now consider the `Boston` housing data set, from the `ISLP` library.

(a) Based on this data set, provide an estimate for the population mean of `medv`. Call this estimate $\hat{\mu}$.

(b) Provide an estimate of the standard error of $\hat{\mu}$. Interpret this result.
*Hint: We can compute the standard error of the sample mean by dividing the sample standard deviation by the square root of the number of observations.*

(c) Now estimate the standard error of $\hat{\mu}$ using the bootstrap. How does this compare to your answer from (b)?

(d) Based on your bootstrap estimate from (c), provide a 95 % confidence interval for the mean of `medv`. Compare it to the results obtained by using `Boston['medv'].std()` and the two standard error rule (3.9).
*Hint: You can approximate a 95 % confidence interval using the formula* $[\hat{\mu}-2\mathrm{SE}(\hat{\mu}), \hat{\mu}+2\mathrm{SE}(\hat{\mu})]$.

(e) Based on this data set, provide an estimate, $\hat{\mu}_{med}$, for the median value of `medv` in the population.

(f) We now would like to estimate the standard error of $\hat{\mu}_{med}$. Unfortunately, there is no simple formula for computing the standard error of the median. Instead, estimate the standard error of the median using the bootstrap. Comment on your findings.

(g) Based on this data set, provide an estimate for the tenth percentile of `medv` in Boston census tracts. Call this quantity $\hat{\mu}_{0.1}$. (You can use the `np.percentile()` function.)

(h) Use the bootstrap to estimate the standard error of $\hat{\mu}_{0.1}$. Comment on your findings.

---

**Solution.**

**Notation.**

- $x_i$: `medv` of tract $i$, median value of owner-occupied homes in thousands of US dollars; $N = 506$ tracts (file `Listas/data/Boston.csv`, from the ISLP repository).
- $\hat{\mu}$: sample mean; $s$: sample standard deviation (divisor $N-1$); SE: standard error.
- $B = 10{,}000$ bootstrap samples $x^{*b}$ of size $N$ drawn with replacement (NumPy seed 1); $\hat{\mu}^{*b}$, $\hat{\mu}^{*b}_{med}$, $\hat{\mu}^{*b}_{0.1}$: the statistic on sample $b$.

**Step — (a) Mean.**

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} x_i = 22.5328. \tag{7.1}$$

**Step — (b) Formula standard error.**

$$\mathrm{SE}(\hat{\mu}) = \frac{s}{\sqrt{N}}. \tag{7.2}$$

Substituting into (7.2):

$$\mathrm{SE}(\hat{\mu}) = \frac{9.1971}{\sqrt{506}} = \frac{9.1971}{22.4944} = 0.4089.$$

If we could draw many samples of 506 tracts, their means would scatter around the population mean with a standard deviation of about 0.41 (roughly 409 US dollars).

**Step — (c) Bootstrap standard error.**

$$\mathrm{SE}_B(\hat{\mu}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\mu}^{*b} - \frac{1}{B}\sum_{b'=1}^{B}\hat{\mu}^{*b'}\right)^2} = 0.4098. \tag{7.3}$$

It agrees with (7.2) to within 0.0009, the size of the Monte Carlo error of $B = 10{,}000$ replicates: for the mean, the bootstrap reproduces the textbook formula.

**Step — (d) 95% intervals.** Using the two-standard-error rule with (7.3):

$$[\hat{\mu} - 2\,\mathrm{SE},\ \hat{\mu} + 2\,\mathrm{SE}]. \tag{7.4}$$

Substituting into (7.4):

$$\begin{aligned}&[\,22.5328 - 2(0.4098),\ 22.5328 + 2(0.4098)\,]\\ &= [\,22.5328 - 0.8196,\ 22.5328 + 0.8196\,] = [\,21.7132,\ 23.3524\,].\end{aligned}$$

With `Boston['medv'].std()` $= s$ in (7.2): $[\,22.5328{-}0.8177,\ 22.5328{+}0.8177\,] = [\,21.7151,\ 23.3505\,]$. The two intervals differ only in the third decimal.

**Step — (e) Median.** With $N$ even, $\hat{\mu}_{med}$ is the average of the 253rd and 254th ordered values: $\hat{\mu}_{med} = 21.2$.

**Step — (f) Bootstrap SE of the median.** Applying (7.3) to $\hat{\mu}^{*b}_{med}$: $\mathrm{SE}_B(\hat{\mu}_{med}) = 0.3796$. The median is estimated about as precisely as the mean, even a little more: `medv` is right-skewed and capped at 50, and the median ignores the tail observations that inflate the variance of the mean.

**Step — (g) Tenth percentile.** `np.percentile(medv, 10)` (linear interpolation) gives $\hat{\mu}_{0.1} = 12.75$.

**Step — (h) Bootstrap SE of the tenth percentile.** Applying (7.3) to $\hat{\mu}^{*b}_{0.1}$: $\mathrm{SE}_B(\hat{\mu}_{0.1}) = 0.5080$, larger than for the median. Fewer observations lie near the 10th percentile, so it moves more from sample to sample; asymptotically $\mathrm{SE}(\hat{q}_p) \approx \sqrt{p(1-p)/N}/f(q_p)$, which grows where the density $f$ is low.

**Step — Python.**

```
medv = pd.read_csv("Listas/data/Boston.csv")["medv"].to_numpy()   # N = 506
mu, se = medv.mean(), medv.std(ddof=1) / np.sqrt(len(medv))       # 22.5328, 0.4089
rng = np.random.default_rng(1)
boot = medv[rng.integers(0, 506, size=(10000, 506))]              # B = 10,000 samples
boot.mean(1).std(ddof=1)                      # 0.4098
np.median(boot, 1).std(ddof=1)                # 0.3796   (median 21.2)
np.percentile(boot, 10, axis=1).std(ddof=1)   # 0.5080   (10th percentile 12.75)
```

*Intuition.* The bootstrap replaces a formula by resampling: for the mean it reproduces $s/\sqrt{N}$, and for the median and percentiles, which have no simple formula, it is the practical tool. The SEs also show that tail quantiles are noisier than central ones.

**Ref:** ISLP (2023), §5.2 (the bootstrap, pp. 212–215) and §3.1.2 (the two standard error rule (3.9)).

## Block 5 — Bolfarine & Sandoval, *Introdução à Inferência Estatística* *(B 1.4, 1.9, 1.12 → 2.2, 2.4, 2.10 → 3.6, 3.13, 3.17)*

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**Question 8.** B 1.4 — MSE of $S^2$ and $\hat{\sigma}^2$.

**Source:** Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §1.4 Exercícios, Exercício 1.4, p. 13 — reproduced verbatim. The expressions it refers to are reproduced verbatim from Exemplo 1.3.6, p. 9.

**1.4.** Verifique a validade das expressões (1.3.10) e (1.3.11).

*Expressões referidas (Exemplo 1.3.6, p. 9):* "Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim N(\mu, \sigma^2)$. Conforme visto no Exemplo 1.3.3, $\hat{\sigma}^2$ é um estimador viciado para $\sigma^2$. De (1.3.3) segue que

$$S^2 = \frac{n}{n-1}\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

é um estimador não viciado para $\sigma^2$. Por outro lado, temos (ver Exercício 1.4) que

$$EQM[S^2] = Var[S^2] = \frac{2\sigma^4}{n-1}, \tag{1.3.10}$$

e que

$$EQM[\hat{\sigma}^2] = \frac{2\sigma^4}{(n-1)}\left[1 - \frac{(3n-1)}{2n^2}\right]." \tag{1.3.11}$$

---

**Solution.**

**Notation.**

- $X_1, \ldots, X_n$ i.i.d. $N(\mu, \sigma^2)$, $n \geq 2$; $\overline{X}$: sample mean.
- $\hat{\sigma}^2 = \frac{1}{n}\sum_i (X_i - \overline{X})^2$ (MLE, Exemplo 1.3.3); $S^2 = \frac{1}{n-1}\sum_i (X_i - \overline{X})^2$.
- $EQM[T] = \mathbb{E}[(T - \sigma^2)^2]$: mean squared error; $\chi^2_\nu$: chi-square with $\nu$ degrees of freedom.

**Step — Sampling distribution.** For a normal sample (Student–Fisher theorem),

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad \mathbb{E}(\chi^2_\nu) = \nu, \quad \mathrm{Var}(\chi^2_\nu) = 2\nu. \tag{8.1}$$

**Step — Moments of $S^2$.** Writing $S^2 = \frac{\sigma^2}{n-1} \cdot \frac{(n-1)S^2}{\sigma^2}$ and using (8.1):

$$\mathbb{E}[S^2] = \frac{\sigma^2}{n-1}(n-1) = \sigma^2, \tag{8.2}$$

$$\mathrm{Var}[S^2] = \frac{\sigma^4}{(n-1)^2}(2(n-1)) = \frac{2\sigma^4}{n-1}. \tag{8.3}$$

**Step — (1.3.10).** The decomposition

$$EQM[T] = \mathrm{Var}[T] + \big(\mathbb{E}[T] - \sigma^2\big)^2 \tag{8.4}$$

with (8.2) (zero bias) and (8.3) substituted gives $EQM[S^2] = \frac{2\sigma^4}{n-1} + (0)^2 = \frac{2\sigma^4}{n-1}$, which is (1.3.10).

**Step — Moments of $\hat{\sigma}^2$.** Since

$$\hat{\sigma}^2 = \frac{n-1}{n}S^2, \tag{8.5}$$

substituting (8.2) and (8.3):

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}(\sigma^2) \implies \text{bias} = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}, \tag{8.6}$$

$$\mathrm{Var}[\hat{\sigma}^2] = \frac{(n-1)^2}{n^2}\Big(\frac{2\sigma^4}{n-1}\Big) = \frac{2(n-1)\sigma^4}{n^2}. \tag{8.7}$$

**Step — (1.3.11).** Substituting (8.6) and (8.7) into (8.4):

$$EQM[\hat{\sigma}^2] = \frac{2(n-1)\sigma^4}{n^2} + \Big(-\frac{\sigma^2}{n}\Big)^2 = \frac{(2n-1)\sigma^4}{n^2}. \tag{8.8}$$

Expanding the book's form:

$$\frac{2\sigma^4}{n-1}\left[1 - \frac{3n-1}{2n^2}\right] = \frac{2\sigma^4}{n-1} \cdot \frac{2n^2 - 3n + 1}{2n^2} = \frac{2\sigma^4}{n-1} \cdot \frac{(2n-1)(n-1)}{2n^2} = \frac{(2n-1)\sigma^4}{n^2},$$

equal to (8.8), which verifies (1.3.11).

**Step — Comparison.** The bracket in (1.3.11) is below 1 for every $n \geq 1$, so $EQM[\hat{\sigma}^2] < EQM[S^2]$. For $n = 10$: $EQM[S^2] = 2/9\,\sigma^4 = 0.2222\,\sigma^4$ and $EQM[\hat{\sigma}^2] = 19/100\,\sigma^4 = 0.19\,\sigma^4$.

*Intuition.* Multiplying $S^2$ by $(n-1)/n < 1$ adds a small downward bias but cuts the variance by more, so the biased MLE wins on MSE. Unbiasedness is one criterion among several, not a guarantee of accuracy.

**Ref:** Bolfarine & Sandoval (2010), §1.3, Exemplos 1.3.3 and 1.3.6 (pp. 6–9).

**Question 9.** B 1.9 — Bernoulli: unbiased $X$ versus the constant $1/2$.
**Source:** Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §1.4 Exercícios, Exercício 1.9, p. 14 — reproduced verbatim.

**1.9.** Seja $X$ uma única variável aleatória com distribuição de Bernoulli com parâmetro $\theta$. Sejam $\hat{\theta}_1 = X$ e $\hat{\theta}_2 = 1/2$ dois estimadores de $\theta$.
(i) Verifique se $\hat{\theta}_1$ e $\hat{\theta}_2$ são não viciados para $\theta$.
(ii) Compare os $EQMs$. Faça um gráfico dos $EQMs$ como função de $\theta$.

---

**Solution.**

**Notation.**

- $X \sim \text{Bernoulli}(\theta)$, $\theta \in (0,1)$: $P(X=1)=\theta$, $\mathbb{E}X=\theta$, $\text{Var}\,X=\theta(1-\theta)$.
- $EQM_j(\theta) = \mathbb{E}[(\hat{\theta}_j - \theta)^2]$, $j = 1, 2$.

**Step — (i) Bias.**

$$\mathbb{E}[\hat{\theta}_1] = \mathbb{E}[X] = \theta, \qquad \mathbb{E}[\hat{\theta}_2] = \tfrac{1}{2}. \tag{9.1}$$

$\hat{\theta}_1$ is unbiased; $\hat{\theta}_2$ has bias $\frac{1}{2} - \theta$, so it is unbiased only at $\theta = 1/2$.

**Step — (ii) MSEs.** Using $EQM = \text{Var} + \text{bias}^2$ with (9.1):

$$EQM_1(\theta) = \theta(1-\theta) + 0^2 = \theta(1-\theta), \tag{9.2}$$

$$EQM_2(\theta) = 0 + \left(\tfrac{1}{2} - \theta\right)^2 = \left(\tfrac{1}{2} - \theta\right)^2. \tag{9.3}$$

**Step — (ii) Where each wins.** $EQM_2 < EQM_1$ means, by (9.2)–(9.3),

$$\tfrac{1}{4} - \theta + \theta^2 < \theta - \theta^2.$$

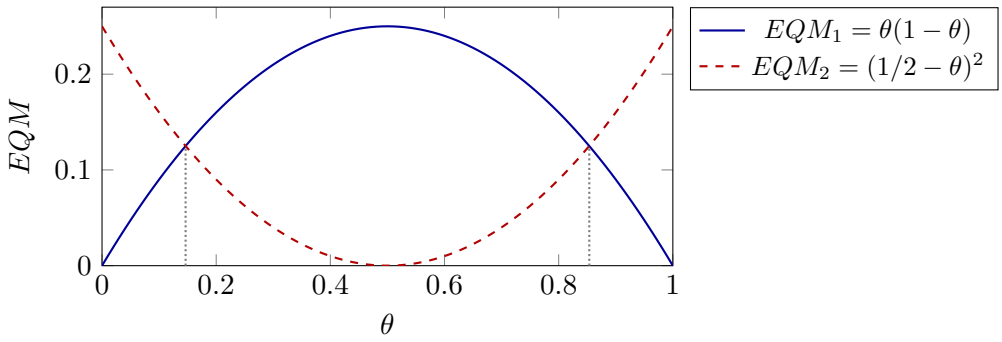
Collecting terms:

$$2\theta^2 - 2\theta + \tfrac{1}{4} < 0. \tag{9.4}$$

The roots of (9.4) are

$$\theta = \frac{2 \pm \sqrt{4 - 4 \cdot 2 \cdot \frac{1}{4}}}{2 \cdot 2} = \frac{2 \pm \sqrt{2}}{4}, \quad \text{i.e.} \quad 0.1464 \text{ and } 0.8536.$$

So $\hat{\theta}_2 = 1/2$ has smaller MSE for $\theta \in (0.1464,\, 0.8536)$ and $\hat{\theta}_1 = X$ for $\theta$ outside it; the MSEs are equal at the two roots.




*Intuition.* A constant guess has zero variance and is excellent when it happens to be near the truth, and poor otherwise. Neither estimator has smaller MSE for every $\theta$, so MSE alone cannot rank them; this is why the theory restricts the class (unbiased estimators) before looking for a best one.

**Ref:** Bolfarine & Sandoval (2010), §1.3 (erro quadrático médio, pp. 4–13).

**Question 10.** B 1.12 — $U(0,\theta)$: $c_1\overline{X}$ versus $c_2 X_{(n)}$.
**Source:** Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §1.4 Exercícios, Exercício 1.12, p. 14 — reproduced verbatim.

**1.12.** Sejam $X_1, \ldots, X_n$ uma amostra aleatória de tamanho $n$ da distribuição de uma variável aleatória $X \sim U(0,\theta)$. Considere os estimadores $\hat{\theta}_1 = c_1\overline{X}$ e $\hat{\theta}_2 = c_2 X_{(n)}$.
(i) Encontre $c_1$ e $c_2$ que tornam os estimadores não viciados.
(ii) Encontre e compare os $EQMs$ dos dois estimadores.

---

**Solution.**

**Notation.**

- $X_i$ i.i.d. $U(0,\theta)$, $\theta > 0$: $\mathbb{E}X = \theta/2$, $\operatorname{Var} X = \theta^2/12$.
- $X_{(n)} = \max_i X_i$; $c_1, c_2 > 0$ constants; $EQM_j = \mathbb{E}[(\hat{\theta}_j - \theta)^2]$.

**Step — Distribution of $X_{(n)}$.** $X_{(n)} \le x$ iff every $X_i \le x$, so for $0 \le x \le \theta$

$$F_{(n)}(x) = \Big(\frac{x}{\theta}\Big)^n, \qquad f_{(n)}(x) = \frac{n\,x^{n-1}}{\theta^n} \tag{10.1}$$

(Bolfarine's (1.3.14)). Its moments follow from (10.1):

$$\mathbb{E}[X_{(n)}^r] = \int_0^\theta x^r \frac{n x^{n-1}}{\theta^n} dx = \frac{n\,\theta^r}{n+r}. \tag{10.2}$$

Setting $r = 1, 2$ in (10.2):

$$\mathbb{E}[X_{(n)}] = \frac{n\theta}{n+1}, \qquad \mathbb{E}[X_{(n)}^2] = \frac{n\theta^2}{n+2}. \tag{10.3}$$

Hence

$$\operatorname{Var}[X_{(n)}] = \frac{n\theta^2}{n+2} - \Big(\frac{n\theta}{n+1}\Big)^2 = \frac{n\theta^2\,[(n+1)^2 - n(n+2)]}{(n+1)^2(n+2)} = \frac{n\theta^2}{(n+1)^2(n+2)} \tag{10.4}$$

(Bolfarine's (1.3.15)).

**Step — (i) Unbiasing constants.** $\mathbb{E}[c_1\overline{X}] = c_1\theta/2 = \theta$ and, by (10.3), $\mathbb{E}[c_2 X_{(n)}] = c_2\, n\theta/(n+1) = \theta$:

$$c_1 = 2, \qquad c_2 = \frac{n+1}{n}. \tag{10.5}$$

**Step — (ii) MSE of $\hat{\theta}_1$.** Both are unbiased, so $EQM = \operatorname{Var}$. With $\operatorname{Var}\overline{X} = \theta^2/(12n)$:

$$EQM_1 = \operatorname{Var}(2\overline{X}) = 4 \cdot \frac{\theta^2}{12n} = \frac{\theta^2}{3n}. \tag{10.6}$$

**Step — (ii) MSE of $\hat{\theta}_2$.** Substituting (10.5) and (10.4):

$$EQM_2 = \Big(\frac{n+1}{n}\Big)^2 \cdot \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{\theta^2}{n(n+2)}. \tag{10.7}$$

**Step — (ii) Comparison.** Dividing (10.7) by (10.6):

$$\frac{EQM_2}{EQM_1} = \frac{\theta^2/[n(n+2)]}{\theta^2/(3n)} = \frac{3}{n+2}, \tag{10.8}$$

which is 1 at $n = 1$ (both estimators are $2X_1$) and below 1 for every $n \ge 2$. For $n = 10$, $\theta = 1$: $EQM_1 = 1/30 = 0.0333$, $EQM_2 = 1/120 = 0.0083$, ratio $3/12 = 0.25$. A Monte Carlo run with 200,000 samples (check script) gives 0.0333 and 0.0084.

*Intuition.* $X_{(n)}$ is sufficient for $\theta$ and sits right below the upper end of the support, which is exactly what we estimate; $\overline{X}$ throws that information away. Because the support depends on $\theta$, the MSE of $\hat{\theta}_2$ falls like $1/n^2$, faster than the $1/n$ of regular problems (Cramér–Rao does not apply).

**Ref:** Bolfarine & Sandoval (2010), §1.3, Exemplo 1.3.8 and (1.3.14)–(1.3.15) (pp. 10–11).

**Question 11.** B 2.2 — Binomial$(2,\theta)$: Cramér–Rao bound and an efficient estimator.
**Source:** Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §2.6 Exercícios, Exercício 2.2, p. 32 — reproduced verbatim.

**2.2.** Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim \mathit{Binomial}(2,\theta)$.
(i) Encontre o limite inferior da variância dos estimadores não viciados de $\theta$.
(ii) Encontre uma estatística suficiente para $\theta$.
(iii) Obtenha um estimador não viciado para $\theta$ que seja função da estatística suficiente.
(iv) Verifique se o estimador é eficiente.

---

**Solution.**

**Notation.**

- $f(x \mid \theta) = \binom{2}{x}\theta^x(1-\theta)^{2-x}$, $x \in \{0,1,2\}$, $\theta \in (0,1)$; $\mathbb{E}X = 2\theta$, $\operatorname{Var} X = 2\theta(1-\theta)$.
- $LI(\theta)$: Cramér–Rao lower bound, Bolfarine (2.1.1); $T = \sum_i X_i \sim \text{Binomial}(2n,\theta)$.

**Step — (i) Score.**

$$\log f(x \mid \theta) = \log\tbinom{2}{x} + x\log\theta + (2-x)\log(1-\theta). \tag{11.1}$$

Differentiating (11.1):

$$\frac{\partial \log f}{\partial \theta} = \frac{x}{\theta} - \frac{2-x}{1-\theta} = \frac{x(1-\theta)-(2-x)\theta}{\theta(1-\theta)}.$$

Simplifying the numerator:

$$\frac{\partial \log f}{\partial \theta} = \frac{x-2\theta}{\theta(1-\theta)}. \tag{11.2}$$

**Step — (i) Information and bound.** Squaring (11.2) and taking expectations, with $\mathbb{E}(X-2\theta)^2 = \operatorname{Var} X$:

$$\mathbb{E}\Big[\Big(\frac{\partial \log f(X \mid \theta)}{\partial \theta}\Big)^2\Big] = \frac{2\theta(1-\theta)}{\theta^2(1-\theta)^2} = \frac{2}{\theta(1-\theta)}. \tag{11.3}$$

Substituting (11.3) into Bolfarine's (2.1.1), $LI(\theta) = 1/\{nE[(\partial \log f/\partial\theta)^2]\}$:

$$LI(\theta) = \frac{1}{n \cdot \frac{2}{\theta(1-\theta)}} = \frac{\theta(1-\theta)}{2n}. \tag{11.4}$$

**Step — (ii) Sufficient statistic.** The joint probability function is

$$\prod_{i=1}^{n} f(x_i \mid \theta) = \underbrace{\prod_{i=1}^{n}\tbinom{2}{x_i}}_{h(\mathbf{x})}\ \underbrace{\theta^{\sum x_i}(1-\theta)^{2n-\sum x_i}}_{g_\theta(T(\mathbf{x}))}, \tag{11.5}$$

so by the factorisation criterion (Teorema 2.2.1) $T = \sum_{i=1}^n X_i$ is sufficient.

**Step — (iii) Unbiased function of $T$.** $\mathbb{E}T = n\,\mathbb{E}X = 2n\theta$, so

$$\hat\theta = \frac{T}{2n} = \frac{\overline{X}}{2}, \qquad \mathbb{E}\hat\theta = \frac{2n\theta}{2n} = \theta. \tag{11.6}$$

**Step — (iv) Efficiency.** With $\operatorname{Var} T = 2n\theta(1-\theta)$:

$$\operatorname{Var}\hat\theta = \frac{\operatorname{Var} T}{4n^2} = \frac{2n\theta(1-\theta)}{4n^2} = \frac{\theta(1-\theta)}{2n}. \tag{11.7}$$

(11.7) equals (11.4), so $e(\hat\theta) = LI(\theta)/\operatorname{Var}\hat\theta = 1$: $\hat\theta$ is efficient (and, $T$ being complete, also the ENVVUM).

*Intuition.* A Binomial$(2,\theta)$ observation is two Bernoulli trials, so the sample is $2n$ trials and the natural estimator is the success proportion $T/(2n)$. The score (11.2) is linear in $x$, which is the condition for the bound to be attained.

**Ref:** Bolfarine & Sandoval (2010), §2.1 (Teorema 2.1.1, pp. 15–19) and §2.2 (Teorema 2.2.1, pp. 20–23).

**Question 12.** B 2.4 — $X_1 + 2X_2$ is not sufficient for a Poisson mean.
**Source:** Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §2.6 Exercícios, Exercício 2.4, p. 32 — reproduced verbatim.

**2.4.** Sejam $X_1, X_2$ uma amostra aleatória da variável aleatória $X \sim Poisson(\theta)$. Mostre que $T = X_1 + 2X_2$ não é suficiente para $\theta$.

---

**Solution.**

**Notation.**

- $X_1, X_2$ i.i.d. Poisson($\theta$), $P(X = x) = \theta^x e^{-\theta}/x!$, $\theta > 0$; $T = X_1 + 2X_2$.
- Sufficiency (Definição 2.2.1): $P(\mathbf{X} = \mathbf{x} \mid T = t)$ does not depend on $\theta$.

**Step — Pick a value of $T$.** The pairs with $T = 2$ are

$$\{T = 2\} = \{(X_1, X_2) = (2, 0)\} \cup \{(X_1, X_2) = (0, 1)\}. \tag{12.1}$$

**Step — Probabilities.** By independence,

$$P(X_1 = 2, X_2 = 0) = \frac{\theta^2 e^{-\theta}}{2!} \cdot e^{-\theta} = \frac{\theta^2 e^{-2\theta}}{2}, \tag{12.2}$$

$$P(X_1 = 0, X_2 = 1) = e^{-\theta} \cdot \theta e^{-\theta} = \theta e^{-2\theta}. \tag{12.3}$$

Adding (12.2) and (12.3) as (12.1) requires:

$$P(T = 2) = e^{-2\theta}\Big(\frac{\theta^2}{2} + \theta\Big). \tag{12.4}$$

**Step — Conditional probability.** Dividing (12.2) by (12.4):

$$P(X_1 = 2, X_2 = 0 \mid T = 2) = \frac{\theta^2 e^{-2\theta}/2}{e^{-2\theta}(\theta^2/2 + \theta)} = \frac{\theta^2/2}{\theta^2/2 + \theta}.$$

Dividing numerator and denominator by $\theta/2$:

$$P(X_1 = 2, X_2 = 0 \mid T = 2) = \frac{\theta}{\theta + 2}, \tag{12.5}$$

which depends on $\theta$ (e.g. $1/3 = 0.3333$ at $\theta = 1$, $2/4 = 0.5$ at $\theta = 2$). So $T$ is not sufficient.

*Intuition.* The sufficient statistic here is $X_1 + X_2$. Weighting $X_2$ twice merges sample points with different totals: $T = 2$ mixes "two events" with "one event", and the split between these cases still carries information about $\theta$.

**Ref:** Bolfarine & Sandoval (2010), §2.2 (estatísticas suficientes, pp. 20–23; Exemplo 2.2.4).

**Question 13.** B 2.10 — ENVVUM of $\theta(1-\theta)$ for Bernoulli data.
**Source:** Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §2.6 Exercícios, Exercício 2.10, p. 33 — reproduced verbatim.

**2.10.** Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória. $X \sim Bernoulli(\theta)$. Obtenha o ENVVUM para $\theta(1-\theta)$.
Sugestão: verifique se $S^2 = \frac{n}{(n-1)}\overline{X}(1-\overline{X})$ é não viciado para $\theta(1-\theta)$.

---

**Solution.**

**Notation.**

- $X_i$ i.i.d. Bernoulli($\theta$), $\theta \in (0,1)$, $n \geq 2$; $T = \sum_i X_i \sim$ Binomial$(n, \theta)$; $\overline{X} = T/n$.
- ENVVUM: *estimador não viciado de variância uniformemente mínima* (UMVUE).

**Step — Moments of $\overline{X}$.**

$$\mathbb{E}\overline{X} = \theta, \qquad \operatorname{Var}\overline{X} = \frac{\theta(1-\theta)}{n}. \tag{13.1}$$

Hence

$$\mathbb{E}\overline{X}^2 = \operatorname{Var}\overline{X} + (\mathbb{E}\overline{X})^2 = \frac{\theta(1-\theta)}{n} + \theta^2. \tag{13.2}$$

**Step — Bias of the plug-in.** Using (13.1) and (13.2):

$$\mathbb{E}\big[\overline{X}(1-\overline{X})\big] = \mathbb{E}\overline{X} - \mathbb{E}\overline{X}^2 = \theta - \Big(\frac{\theta(1-\theta)}{n} + \theta^2\Big).$$

Since $\theta - \theta^2 = \theta(1-\theta)$:

$$\mathbb{E}\big[\overline{X}(1-\overline{X})\big] = \theta(1-\theta) - \frac{\theta(1-\theta)}{n} = \frac{n-1}{n}\,\theta(1-\theta). \tag{13.3}$$

**Step — Unbiased correction.** Multiplying (13.3) by $n/(n-1)$:

$$\mathbb{E}[S^2] = \frac{n}{n-1}\cdot\frac{n-1}{n}\,\theta(1-\theta) = \theta(1-\theta). \tag{13.4}$$

**Step — Function of a complete sufficient statistic.** Substituting $\overline{X} = T/n$:

$$S^2 = \frac{n}{n-1}\cdot\frac{T}{n}\Big(1-\frac{T}{n}\Big) = \frac{T(n-T)}{n(n-1)}. \tag{13.5}$$

$T$ is sufficient and complete for the Bernoulli family (one-parameter exponential family, Teorema 2.5.2). By Lehmann–Scheffé (Teorema 2.5.3), the unbiased estimator (13.4) that is a function of $T$, (13.5), is the unique ENVVUM of $\theta(1-\theta)$.

*Intuition.* Since $X_i^2 = X_i$, $S^2$ is exactly the usual sample variance with divisor $n-1$. The plug-in $\overline{X}(1-\overline{X})$ is the MLE of $\theta(1-\theta)$ (Exemplo 3.2.1) and is biased downward by the factor $(n-1)/n$, the same correction as $\hat{\sigma}^2$ versus $S^2$ in B 1.4.

**Ref:** Bolfarine & Sandoval (2010), §2.5 (completude, Teoremas 2.5.2–2.5.3, pp. 29–32).

**Question 14.** B 3.6 — MLE of $\mu^2$ versus the unbiased $\overline{X}^2 - 1/n$.
**Source:** Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §3.8 Exercícios, Exercício 3.6, p. 53 — reproduced verbatim. Exercise 2.9, to which it refers, is reproduced verbatim from §2.6, p. 33.

**3.6.** Encontre o estimador de máxima verossimilhança de $\theta^2$ no Exercício 2.9 e compare seu erro quadrático médio com o do estimador eficiente $\hat{\gamma}$ dado no Exercício 2.9, (i).

*Exercício referido (p. 33):* **2.9.** Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim N(\mu, 1)$.
(i) Mostre que o estimador $\hat{\gamma} = \overline{X}^2 - 1/n$ é não viciado para $g(\mu) = \mu^2$.
(ii) Existe ENVVUM para $\mu^2$?
(iii) Encontre o limite inferior da variância dos estimadores não viciados de $g(\mu) = \mu^2$ e verifique se $\hat{\gamma}$ é eficiente.
*Note:* the parameter of Exercise 2.9 is $\mu$; the "$\theta^2$" of 3.6 is read as $\mu^2$.

---

**Solution.**

**Notation.**

- $X_i$ i.i.d. $N(\mu, 1)$, $\mu \in \mathbb{R}$; $\overline{X} \sim N(\mu, 1/n)$; $Z \sim N(0,1)$.
- $\widehat{\mu^2}$: MLE of $\mu^2$; $\hat{\gamma} = \overline{X}^2 - 1/n$; $EQM(T) = \mathbb{E}[(T-\mu^2)^2]$.

**Step — MLE.** The log-likelihood $\ell(\mu) = -\frac{n}{2}\log 2\pi - \frac{1}{2}\sum_i (x_i-\mu)^2$ has score $\sum_i (x_i-\mu) = 0$, so $\hat\mu = \overline{X}$. The map $\mu \mapsto \mu^2$ is not one-to-one, but the likelihood of $\eta = \mu^2$, $L^*(\eta) = \max\{L(\sqrt{\eta}), L(-\sqrt{\eta})\}$, is maximised at $\eta = \overline{X}^2$ (the extension of the invariance principle, Teorema 3.2.2):

$$\widehat{\mu^2} = \overline{X}^2. \tag{14.1}$$

**Step — Moments of $\overline{X}^2$.** Write

$$\overline{X} = \mu + \frac{Z}{\sqrt{n}} \implies \overline{X}^2 = \mu^2 + \frac{2\mu Z}{\sqrt{n}} + \frac{Z^2}{n}. \tag{14.2}$$

With $\mathbb{E}Z = 0$, $\mathbb{E}Z^2 = 1$, $\operatorname{Var} Z^2 = 2$, $\operatorname{Cov}(Z, Z^2) = \mathbb{E}Z^3 = 0$, (14.2) gives

$$\mathbb{E}\overline{X}^2 = \mu^2 + \frac{1}{n}, \tag{14.3}$$

$$\operatorname{Var}\overline{X}^2 = \frac{4\mu^2}{n}\operatorname{Var} Z + \frac{1}{n^2}\operatorname{Var} Z^2 = \frac{4\mu^2}{n} + \frac{2}{n^2}. \tag{14.4}$$

**Step — MSE of the MLE.** By (14.3) the bias is $1/n$; adding its square to (14.4):

$$EQM(\overline{X}^2) = \frac{4\mu^2}{n} + \frac{2}{n^2} + \Big(\frac{1}{n}\Big)^2 = \frac{4\mu^2}{n} + \frac{3}{n^2}. \tag{14.5}$$

**Step — MSE of $\hat\gamma$.** By (14.3), $\mathbb{E}\hat\gamma = \mu^2$; subtracting a constant leaves the variance (14.4) unchanged:

$$EQM(\hat\gamma) = \operatorname{Var}\hat\gamma = \frac{4\mu^2}{n} + \frac{2}{n^2}. \tag{14.6}$$

**Step — Comparison.** Subtracting (14.6) from (14.5):

$$EQM(\overline{X}^2) - EQM(\hat\gamma) = \frac{3}{n^2} - \frac{2}{n^2} = \frac{1}{n^2} > 0,$$

so $\hat\gamma$ has smaller MSE for every $\mu$ and $n$. For $\mu = 1$, $n = 5$: 0.92 versus 0.88 (Monte Carlo with 400,000 replicates reproduces both).

**Step — A remark on "eficiente".** $\hat\gamma$ is a function of the complete sufficient statistic $\overline{X}$ and unbiased, so it is the ENVVUM (the answer to 2.9(ii)). It is *not* efficient in the Cramér–Rao sense: the bound for $g(\mu) = \mu^2$ is $(g'(\mu))^2/n = 4\mu^2/n$, strictly below (14.6) (answer to 2.9(iii)). The word "eficiente" in 3.6 should be read as "best unbiased".

*Intuition.* Squaring a noisy $\overline{X}$ adds the noise variance $1/n$ on average, so $\overline{X}^2$ overestimates $\mu^2$; $\hat\gamma$ removes exactly that bias without adding variance. Invariance gives the MLE for free but says nothing about its finite-sample bias. ($\hat\gamma$ can be negative; $\max\{\hat\gamma, 0\}$ would lower the MSE further.)

**Ref:** Bolfarine & Sandoval (2010), §3.2.1 (invariância, Teorema 3.2.2, p. 43) and §2.1 (pp. 15–19).

**Question 15.** B 3.13 — MLE of $P[X > 1]$ for exponential data and its large-sample distribution.
**Source:** Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §3.8 Exercícios, Exercício 3.13, p. 54 — reproduced verbatim.

**3.13.** Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X$ com distribuição exponencial com parâmetro $\theta$. Encontre o estimador de máxima verossimilhança de $g(\theta) = P[X > 1]$ e sua distribuição aproximada quando $n$ for grande.

---

**Solution.**

**Notation.**

- $f(x \mid \theta) = \theta e^{-\theta x}$, $x > 0$, $\theta > 0$ (Bolfarine's parametrisation, Exemplo 3.2.3): $\mathbb{E}X = 1/\theta$.
- $g(\theta) = P[X > 1] = e^{-\theta}$; $I_F(\theta)$: Fisher information per observation.

**Step — MLE of $\theta$.**

$$\ell(\theta) = n \log \theta - \theta \sum_{i=1}^{n} x_i. \tag{15.1}$$

Differentiating (15.1) and setting the score to zero:

$$\ell'(\theta) = \frac{n}{\theta} - \sum_i x_i = 0 \implies \hat{\theta} = \frac{1}{\overline{X}}, \tag{15.2}$$

a maximum since $\ell''(\theta) = -n/\theta^2 < 0$.

**Step — MLE of $g$.** $g(\theta) = e^{-\theta}$ is one-to-one on $\theta > 0$, so by invariance (Teorema 3.2.2) and (15.2):

$$g(\hat{\theta}) = e^{-1/\overline{X}}. \tag{15.3}$$

**Step — Fisher information.** $\log f = \log \theta - \theta x$, so $\partial^2 \log f / \partial \theta^2 = -1/\theta^2$ and

$$I_F(\theta) = -\mathbb{E}\Big[\frac{\partial^2 \log f(X \mid \theta)}{\partial \theta^2}\Big] = \frac{1}{\theta^2}. \tag{15.4}$$

**Step — Delta method.** $g'(\theta) = -e^{-\theta}$. Substituting it and (15.4) into Bolfarine's (3.2.4), $\sqrt{n}\,(g(\hat{\theta}) - g(\theta)) \overset{a}{\sim} N\big(0, (g'(\theta))^2 / I_F(\theta)\big)$:

$$\sqrt{n}\,(e^{-1/\overline{X}} - e^{-\theta}) \overset{a}{\sim} N\Big(0,\ \frac{(-e^{-\theta})^2}{1/\theta^2}\Big).$$

Simplifying the variance:

$$e^{-1/\overline{X}} \overset{a}{\sim} N\Big(e^{-\theta},\ \frac{\theta^2 e^{-2\theta}}{n}\Big). \tag{15.5}$$

*Intuition.* The MLE of a probability is the probability evaluated at the MLE, and the delta method turns the $1/n$ variance of $\hat{\theta}$ into a variance for $g(\hat{\theta})$ scaled by $g'(\theta)^2$. The model-free alternative, the share of $X_i > 1$, has variance $e^{-\theta}(1 - e^{-\theta})/n$, never smaller than (15.5): using the exponential model buys precision if the model is right.

**Ref:** Bolfarine & Sandoval (2010), §3.2.1–3.2.2, Exemplo 3.2.3 and (3.2.4) (p. 44).

**Question 16.** B 3.17 — consistency of the estimators in five examples.

**Source:** Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §3.8 Exercícios, Exercício 3.17, p. 55 — reproduced verbatim. The list of the examples' estimators below the statement is a summary (not verbatim) of pp. 37, 43, 44 and 51.

**3.17.** Verifique se os estimadores obtidos nos Exemplos 3.1.2, 3.1.3, 3.2.1, 3.2.3 e 3.6.2 são consistentes.

*Estimators in the examples (summary):* 3.1.2, Bernoulli($\theta$): $\hat{\theta} = \overline{X}$; 3.1.3, $U(0, \theta)$: $\hat{\theta} = X_{(n)}$; 3.2.1, Bernoulli($\theta$): $g(\hat{\theta}) = \overline{X}(1 - \overline{X})$ for $\theta(1 - \theta)$; 3.2.3, $Exp(\theta)$ with density $\theta e^{-\theta x}$: $g(\hat{\theta}) = e^{-1/\overline{X}}$ for $P_\theta[X > 1] = e^{-\theta}$; 3.6.2, gamma$(\alpha, \beta)$ with density $\beta^\alpha x^{\alpha-1} e^{-\beta x}/\Gamma(\alpha)$, method of moments: $\hat{\alpha} = \overline{X}^2/\hat{\sigma}^2$, $\hat{\beta} = \overline{X}/\hat{\sigma}^2$, $\hat{\sigma}^2 = \sum_i (X_i - \overline{X})^2/n$.

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**Solution.**

**Notation.**

- Consistency (Definição 3.7.1): $\lim_{n\to\infty} P(|\hat{\theta} - \theta| > \epsilon) = 0$ for every $\epsilon > 0$, written $\hat{\theta} \xrightarrow{p} \theta$.
- $m_2 = \frac{1}{n}\sum_i X_i^2$; $EQM(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$.

**Step — Tools.** Markov's inequality applied to $(\hat{\theta} - \theta)^2$ (Chebyshev, as in Exemplo 3.7.1):

$$P(|\hat{\theta} - \theta| > \epsilon) \le \frac{EQM(\hat{\theta})}{\epsilon^2}, \tag{16.1}$$

so $EQM(\hat{\theta}) \to 0$ implies consistency. And the continuous-mapping theorem (standard; not stated in Bolfarine): if $\hat{\theta} \xrightarrow{p} \theta$ and $h$ is continuous at $\theta$, then

$$h(\hat{\theta}) \xrightarrow{p} h(\theta), \tag{16.2}$$

also for vectors $\hat{\theta}$.

**Step — Exemplo 3.1.2: $\overline{X}$.** Unbiased with $EQM = \operatorname{Var}\overline{X} = \theta(1-\theta)/n \to 0$, so by (16.1) $\overline{X} \xrightarrow{p} \theta$. Consistent.

**Step — Exemplo 3.1.3: $X_{(n)}$.** Since $X_{(n)} \le \theta$, for $0 < \epsilon < \theta$ and by (10.1):

$$P(|X_{(n)} - \theta| > \epsilon) = P(X_{(n)} < \theta - \epsilon) = \Big(\frac{\theta - \epsilon}{\theta}\Big)^n \xrightarrow[n\to\infty]{} 0, \tag{16.3}$$

and for $\epsilon \ge \theta$ the probability is 0. Consistent (although biased, Exemplo 1.3.8).

**Step — Exemplo 3.2.1: $\overline{X}(1-\overline{X})$.** $\overline{X} \xrightarrow{p} \theta$ (Exemplo 3.1.2) and $h(t) = t(1-t)$ is continuous, so by (16.2) $\overline{X}(1-\overline{X}) \xrightarrow{p} \theta(1-\theta)$. Consistent; its bias $-\theta(1-\theta)/n$ from (13.3) vanishes.

**Step — Exemplo 3.2.3: $e^{-1/\overline{X}}$.** $\mathbb{E}X = 1/\theta$ and $\operatorname{Var} X = 1/\theta^2$, so $EQM(\overline{X}) = 1/(n\theta^2) \to 0$ and, by (16.1),

$$\overline{X} \xrightarrow{p} \frac{1}{\theta}. \tag{16.4}$$

$h(t) = e^{-1/t}$ is continuous at $t = 1/\theta > 0$; by (16.2) and (16.4), $e^{-1/\overline{X}} \xrightarrow{p} e^{-\theta}$. Consistent.

**Step — Exemplo 3.6.2: $\hat{\alpha}, \hat{\beta}$.** For the gamma, $\mathbb{E}X = \alpha/\beta$, $\mathbb{E}X^2 = \alpha/\beta^2 + \alpha^2/\beta^2$, and all moments are finite, so by (16.1) (the variances of $\overline{X}$ and $m_2$ are $O(1/n)$):

$$\overline{X} \xrightarrow{p} \frac{\alpha}{\beta}, \qquad m_2 \xrightarrow{p} \frac{\alpha}{\beta^2} + \frac{\alpha^2}{\beta^2}. \tag{16.5}$$

Since $\hat{\sigma}^2 = m_2 - \overline{X}^2$, (16.2) applied to (16.5) gives

$$\hat{\sigma}^2 \xrightarrow{p} \Big(\frac{\alpha}{\beta^2} + \frac{\alpha^2}{\beta^2}\Big) - \Big(\frac{\alpha}{\beta}\Big)^2 = \frac{\alpha}{\beta^2}. \tag{16.6}$$

The maps $(a,b) \mapsto a^2/b$ and $(a,b) \mapsto a/b$ are continuous for $b > 0$; by (16.2) with (16.5) and (16.6):

$$\hat{\alpha} = \frac{\overline{X}^2}{\hat{\sigma}^2} \xrightarrow{p} \frac{(\alpha/\beta)^2}{\alpha/\beta^2} = \alpha, \qquad \hat{\beta} = \frac{\overline{X}}{\hat{\sigma}^2} \xrightarrow{p} \frac{\alpha/\beta}{\alpha/\beta^2} = \beta.$$

Consistent. All five estimators are consistent.

*Intuition.* Consistency is a weak, large-sample property: biased estimators ($X_{(n)}$, $\overline{X}(1-\overline{X})$, $\hat{\alpha}$) can be consistent, and smooth functions of consistent estimators inherit it. Sample moments converge by the law of large numbers, and everything built continuously from them follows.

**Ref:** Bolfarine & Sandoval (2010), §3.7 (estimadores consistentes, Definição 3.7.1 and Exemplo 3.7.1, p. 52); examples on pp. 37, 43, 44, 51.

## Block 6 — Morettin & Singer, *Estatística e Ciência de Dados* *(MS 6.15)*

**Question 17.** MS 6.15 — decomposition of the total sum of squares.
**Source:** Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), §6.7 Exercícios, Exercício 15, p. 240 — reproduced verbatim. The definitions it uses are reproduced verbatim from §6.2, p. 185.

**15)** Mostre que $SQTot = SQRes + SQReg$.

*Definições (§6.2, p. 185):* "[...] em que a **soma de quadrados total** é $SQTot = \sum_{i=1}^{n}(y_i - \overline{y})^2$, a **soma de quadrados dos resíduos** é $SQRes = \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2$ e a **soma de quadrados da regressão** é $SQReg = \sum_{i=1}^{n}(\widehat{y}_i - \overline{y})^2$." (Model: $y_i = \alpha + \beta x_i + e_i$, with $\widehat{y}_i = \widehat{\alpha} + \widehat{\beta} x_i$ fitted by least squares.)

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**Solution.**

**Notation.**

- $(x_i, y_i)$, $i = 1, \ldots, n$; $\widehat{\alpha}, \widehat{\beta}$: least-squares estimates (MS (6.6)–(6.7)); $\widehat{y}_i = \widehat{\alpha} + \widehat{\beta} x_i$: fitted values; $\widehat{e}_i = y_i - \widehat{y}_i$: residuals; $\overline{y}$: sample mean of $y$.
- $Q(\alpha, \beta) = \sum_i (y_i - \alpha - \beta x_i)^2$: least-squares criterion.

**Step — Normal equations.** Setting the derivatives of $Q$ to zero at $(\widehat{\alpha}, \widehat{\beta})$:

$$\frac{\partial Q}{\partial \alpha} = -2 \sum_i (y_i - \widehat{\alpha} - \widehat{\beta} x_i) = 0 \implies \sum_i \widehat{e}_i = 0, \tag{17.1}$$

$$\frac{\partial Q}{\partial \beta} = -2 \sum_i x_i (y_i - \widehat{\alpha} - \widehat{\beta} x_i) = 0 \implies \sum_i x_i \widehat{e}_i = 0. \tag{17.2}$$

**Step — Decompose each deviation.** Adding and subtracting $\widehat{y}_i$:

$$y_i - \overline{y} = \widehat{e}_i + (\widehat{y}_i - \overline{y}). \tag{17.3}$$

Squaring (17.3) and summing over $i$:

$$\sum_i (y_i - \overline{y})^2 = \sum_i \widehat{e}_i^2 + \sum_i (\widehat{y}_i - \overline{y})^2 + 2 \sum_i \widehat{e}_i (\widehat{y}_i - \overline{y}). \tag{17.4}$$

**Step — The cross term vanishes.** Substituting $\widehat{y}_i = \widehat{\alpha} + \widehat{\beta} x_i$ in the cross term:

$$\sum_i \widehat{e}_i (\widehat{y}_i - \overline{y}) = \sum_i \widehat{e}_i \big((\widehat{\alpha} + \widehat{\beta} x_i) - \overline{y}\big) = (\widehat{\alpha} - \overline{y}) \sum_i \widehat{e}_i + \widehat{\beta} \sum_i x_i \widehat{e}_i.$$

Substituting (17.1) and (17.2):

$$\sum_i \widehat{e}_i (\widehat{y}_i - \overline{y}) = (\widehat{\alpha} - \overline{y})\,(0) + \widehat{\beta}\,(0) = 0. \tag{17.5}$$

**Step — Conclusion.** Substituting (17.5) into (17.4):

$$\underbrace{\sum_i (y_i - \overline{y})^2}_{SQTot} = \underbrace{\sum_i \widehat{e}_i^2}_{SQRes} + \underbrace{\sum_i (\widehat{y}_i - \overline{y})^2}_{SQReg}.$$

The same proof covers multiple regression with an intercept: the normal equations $\mathbf{X}^\top \widehat{\mathbf{e}} = \mathbf{0}$ include the column of ones (giving (17.1)) and imply $\widehat{\mathbf{y}}^\top \widehat{\mathbf{e}} = \widehat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \widehat{\mathbf{e}} = 0$. Without an intercept (17.1) fails in general and so does the identity. (The check script verifies it numerically on simulated data.)

*Intuition.* Geometrically, the residual vector is orthogonal to every column of the design, hence to $\widehat{\mathbf{y}} - \overline{y}\mathbf{1}$; the decomposition is Pythagoras' theorem in $\mathbb{R}^n$. It is what makes $R^2 = SQReg/SQTot$ a share between 0 and 1.

**Ref:** Morettin & Singer (2021), §6.2 (pp. 184–185) and Figura 6.34 with the result on p. 227.